# Rapport 22/10

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We consider the space of SPD matrices with unit diagonal. We denote this space as $\text{Sym}_1^+(n)$ and $\text{Sym}^+(n)$ as the space of SPD matrices. As we know that $\text{Sym}^+(n)$ is an embedded manifold of $\text{R}^{n\times n}$ so it's also called a manifold (the manifold now is defined in a general treatment - Chap 8 of Boumal's book).

We will derive a manifold structure for $\text{Sym}_1^+(n)$ via the embedded geometry perpective (a quotient manifold structure could be considered later). We rely mainly on the section 8.14 of Boumal's book. Now we define the following smooth function: $h : \text{Sym}^+(n) \to \mathbb{R}^n$ by

$$h(A) = (A_{11} - 1, A_{22} - 1, ..., A_{nn} - 1)^T$$

Note that $h^{-1}(0) = \text{Sym}_1^+(n)$. We will prove that $\text{rank}Dh(A) = n$. Actually, using Taylor expansion directly on this function is not possible since $\text{Sym}^+(n)$ is not an Euclidean space. Since this function is also well-defined on $\text{Sym}(n)$ (which is an Euclidean space), we define the smooth extension of this function $\bar{h}(A) = (A_{11} - 1, A_{22} - 1, ..., A_{nn} - 1)^T, \forall A \in \text{Sym}(n)$. Then the differential map for this function is easily defined via the Taylor expansion:

$$\bar{h}(A + H) = \bar{h}(A) + (H_{11}, H_{22}, ..., H_{nn})^T$$

This implies that $D\bar{h}(A)[H] = (H_{11}, H_{22}, ..., H_{nn})^T, \forall H \in \text{Sym}(n)$. By restriction, we have that

$$Dh(A)[H] = (H_{11}, H_{22}, ..., H_{nn})^T, \forall H \in \text{Sym}(n)$$

We can deduce the differential map of $h$ by using a suitable retraction and use the generalized Taylor expansion. But it will make the problem more complicated. Then we have

$$\text{rank}Dh(A) = \dim(\text{Im}Dh(A)) = \dim(\mathbb{R}^n) = n$$

This concludes that $\text{Sym}_1^+(n)$ is an embedded manifold of $\text{Sym}^+(n)$ (by theorem 8.75 of Boumal's book). Then the tangent space of $\text{Sym}_1^+(n)$ is

$$T_A\text{Sym}_1^+(n) = \text{Ker}Dh(A) = \{H \in \text{Sym}(n) | \text{diag}(H) = 0_n\}$$

**Riemannian metric, Riemannian gradient:** We could use the Riemannian metric inherited from $\text{Sym}^+(n)$ i.e. Affine invariant. Note that $T_A\text{Sym}_1^+(n) \subset T_A\text{Sym}^+(n)$.

Let $f$ be a smooth function from $\mathrm{Sym}_1^+(n)$ to $\mathbb{R}$. And $\mathrm{Proj}_A$ be the orthogonal projection from $T_A\mathrm{Sym}^+(n)$ to $T_A\mathrm{Sym}_1^+(n)$ with respect to the affine invariant metric. Then let $\bar{f}$ be a smooth extension of $f$ on $\mathrm{Sym}^+(n)$ and we have

$$\mathrm{grad}f(A) = \mathrm{Proj}_A(\mathrm{grad}\bar{f}(A))$$

Let us remind some properties of the orthogonal projection map in this situation (the original definition is the Definition 3.60 but we will generalize this definition into the current situation), let $A \in \mathrm{Sym}_1^+(n)$ then :

- $\mathrm{Im}(\mathrm{Proj}_A) = T_A\mathrm{Sym}_1^+(n)$
- $\mathrm{Proj}_A \circ \mathrm{Proj}_A = \mathrm{Proj}_A$
- $\langle U - \mathrm{Proj}_A(U), V\rangle_A^{AI} = 0, \forall U \in T_A\mathrm{Sym}^+(n), V \in T_A\mathrm{Sym}_1^+(n)$

Denote $\mathrm{Sym}_0(n) = T_A\mathrm{Sym}_1^+(n)$ the symmetric matrices with zero diagonal. We have that

$$\langle U - \mathrm{Proj}_A(U), V\rangle_A^{AI} = \mathrm{Tr}(A^{-1}(U - \mathrm{Proj}_A(U))A^{-1}V) = \langle A^{-1}(U - \mathrm{Proj}_A(U))A^{-1}, V\rangle_F = 0, \forall V \in \mathrm{Sym}_0(n)$$

Therefore, $A^{-1}(U - \mathrm{Proj}_A(U))A^{-1} \in \mathrm{Sym}_0(n)^\perp$ that contains matrices with zero components exception diagonal. More specifically, let $\lambda \in \mathbb{R}^n$, then $A^{-1}(U - \mathrm{Proj}_A(U))A^{-1} = \mathrm{Diag}(\lambda)$ where $\lambda$ to be determined and Diag maps from a vector to a matrix with this vector as the diagonal and others components are zero. So we have

$$U - \mathrm{Proj}_A(U) = A\mathrm{Diag}(\lambda)A \iff \mathrm{Proj}_A(U) = U - A\mathrm{Diag}(\lambda)A$$

The remaining thing to do is to define $\lambda$ that satisfies

$$\mathrm{diag}(\mathrm{Proj}_A(U)) = \mathrm{diag}(U - A\mathrm{Diag}(\lambda)A) = 0_n$$

This etablishes a linear system with $n$ equations and $n$ parameters, so we can solve for $\lambda$ easily. To obtain a closed-form expression, we notice that $A$ is a SPD matrix, we can proceed some computations and get the following linear equation:

$$(A \circ A)\lambda = \mathrm{diag}(U) \implies \lambda = (A \circ A)^{-1}\mathrm{diag}(U)$$

where $\circ$ is elementwise product between 2 matrices. To derive this formulation, it's useful to first take care of some cases with low dimension to understand the rule and generalize this observation. Finally, an explicit form for the projection is

$$\mathrm{Proj}_A(U) = U - A\mathrm{Diag}((A \circ A)^{-1}\mathrm{diag}(U))A$$

Therefore, the Riemannian gradient for the loss function:

$$\mathrm{grad}f(A) = \mathrm{grad}\bar{f}(A) - A\mathrm{Diag}((A \circ A)^{-1}\mathrm{diag}(\mathrm{grad}\bar{f}(A)))A$$

**Exponential map:** Let $A \in \mathrm{Sym}_1^+(n)$ and $V \in T_A\mathrm{Sym}_1^+(n)$, $\gamma_V$ be an open curve on $I$ where $[0,1] \subset I$ and satisfies

$$\gamma_V(0) = A, \quad \gamma_V'(0) = V$$

The exponential map is defined as

$$\mathrm{Exp}(X, V) = \mathrm{Exp}_X(V) = \gamma_V(1)$$

In this section, we will try to derive an (inexact) exponential map corresponding to the Affine Invariant metric. Note that, $\mathrm{Sym}_1^+(n)$ also admits a quotient structure, so we can use this structure to define a new type of affine invariant metric (sometimes called quotient-affine metrics).

Inspired by the work [1], they provide a geodesics connecting 2 points in $\mathrm{Sym}_1^+(n)$. However, their framework depended on the start and the end points, whereas Riemannian gradient descent method depends on the starting point and the direction to move along. We can rely on it to derive a valid retraction: Let $A \in \mathrm{Sym}_1^+(n)$ and $V \in T_A\mathrm{Sym}_1^+(n)$, i.e. $V \in \mathrm{Sym}_0(n)$, then we apply the exponential map as for space of SPD matrices, that is

$$\mathrm{Exp}_A(V) = A^{1/2} \exp(A^{-1/2} V A^{-1/2}) A^{1/2} \in \mathrm{Sym}^+(n)$$

Then we can project this into $\mathrm{Sym}_1^+(n)$. This is a way to define a retraction. Note that, this isn't the exponential map, it's only a valid retraction. To get a closed-form expression for the exponential map, we have to consider the quotient structure of $\mathrm{Sym}_1^+(n)$

## References

[1] Paul David and Weiqing Gu. A riemannian structure for correlation matrices. *Operators and Matrices*, 2019.